SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 6

Time Value of Money I — Single Amounts & Annuity Basics

Money now is worth more than the same money later because it can be invested. This week covers future value, present value, cash-flow patterns, ordinary annuity vs annuity due, and perpetuities — applied in the Project Garuda vs Project Samudra data-centre case.

Material

1 Why timing matters: compounding vs discounting

Cash at different dates can't be compared directly — bring them to one point in time. **Compounding** → future value; **discounting** → present value. Use timelines, calculators or Excel; inflows positive, outflows negative. Patterns: single amount, annuity, mixed stream.

2 Future and present value of a single amount

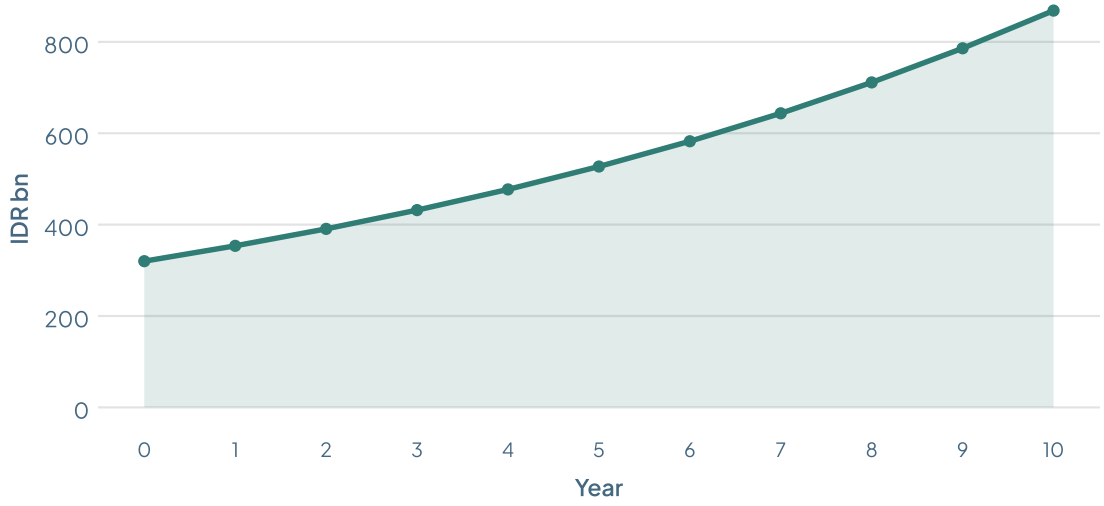
$$FV = PV \times (1 + r)^n; \quad PV = FV \div (1 + r)^n$$

Fred: $100 at 8% → $116.64 after 2 years (compound interest). Paul: $300 in a year at 2% is worth $294.12 today — don't pay more. Pam: $1,700 in 8 years at 4% → $1,242.17. Higher rates and longer horizons raise FV and lower PV.

3 Seeing compounding: the FV and PV curves Deep dive

IDR 320bn at 10.5% grows faster each year (interest on interest).

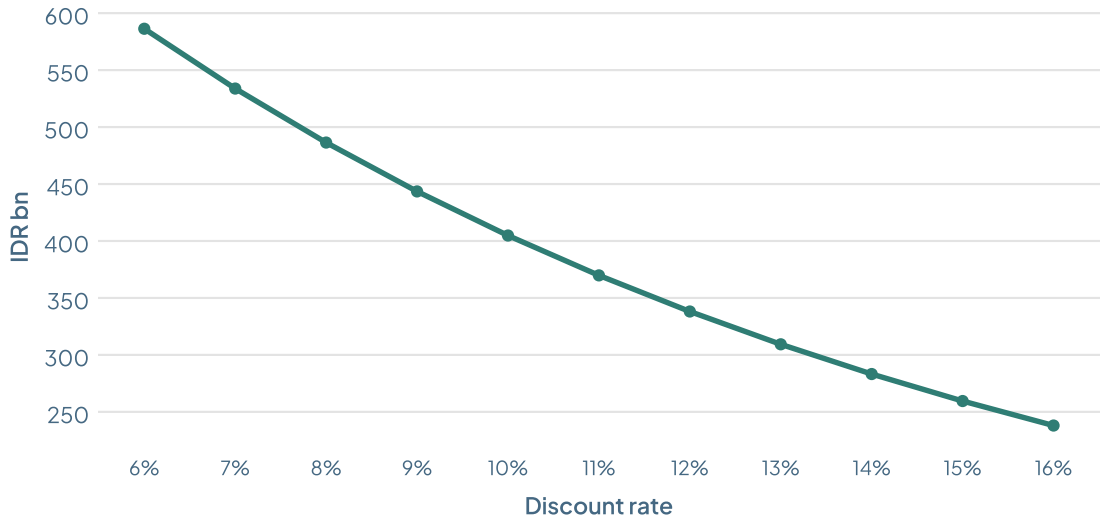
IDR 320bn at 10.5% a year



Year 3: 431.8 · year 5: 527.2 · year 10: 868.5.

The PV of IDR 1,050bn in year 10 shrinks sharply as the discount rate rises.

PV of IDR 1,050bn in year 10 at various rates



From 9% to 14%, PV falls from 443.5 to 283.2.

Rule of 72

72 ÷ 10.5 ≈ 6.9 years to double.

4 Annuities, annuities due and perpetuities

Ordinary annuity: end-of-period payments; **annuity due**: beginning-of-period — always worth more (× (1 + r)). FV ord. = CF[(1+r)^{n}−1]/r; PV ord. = CF[1−(1+r)$^{-n}$]/r. Fran: $1,000/yr for 5 years at 7% → $5,750.74. **Perpetuity** PV = CF ÷ r; growing perpetuity PV = CF$_1$ ÷ (r − g), used for terminal values.

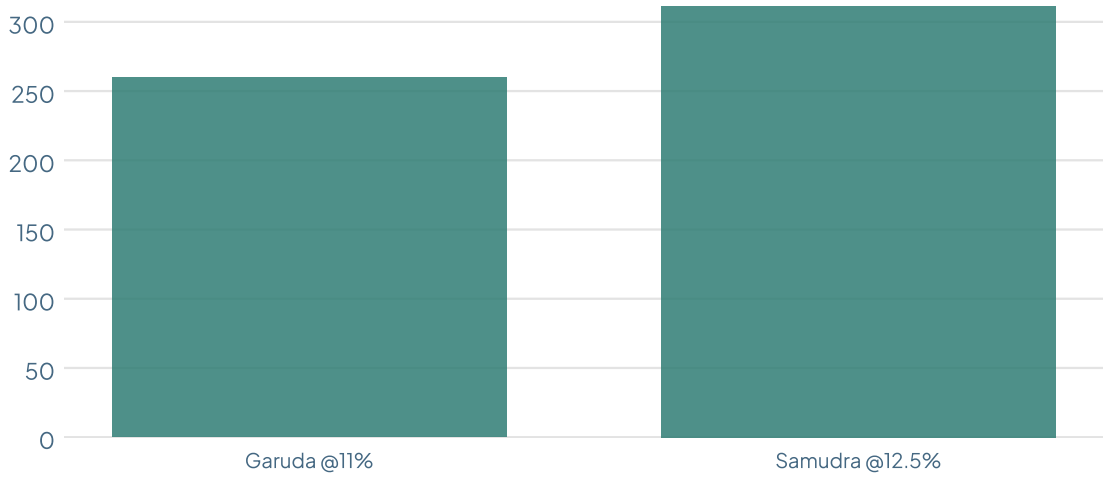
5 Key ideas behind the formulas

- Discount rate = opportunity cost.
- Different risk → different rate.
- Ignore sunk costs; include opportunity costs.
- Compounding frequency matters — compare with EAR = (1 + r/m)^{m} − 1.
- Biggest nominal total ≠ biggest present value.

6 Garuda vs Samudra: comparing properly Deep dive

Convert everything to present value at each project's risk-adjusted rate, then compare NPV.

Project NPV (IDR bn)



From the case discussion; still weigh risk and terminal-value sensitivity.

Do
Risk-adjusted rates, all cash flows, NPV > 0.

Traps
One rate for all, raw totals, sunk costs.

≡ **Supplier G1's implied rate**
540/475 − 1 = 13.7% > 8.75% borrowing cost.

7 Worked numbers: ordinary annuity vs annuity due, and mixed streams

An annuity of $1,000 a year for 5 years at 7% (the slide example):

	Ordinary annuity (end of period)	Annuity due (start of period)
Future value	$5,750.74	$5,750.74 × 1.07 = **$6,153.29**
Present value	$4,100.20	$4,100.20 × 1.07 = **$4,387.21**

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PV of ordinary annuity = CF × [1 - 1/(1+r)ⁿ] ÷ r
PV of annuity due = PV of ordinary annuity × (1 + r)
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Why multiply by (1 + r)? Because each annuity-due payment arrives **one period earlier**, so it compounds (or is discounted) for one period less. That is why an annuity due is always worth more.

The "long method" (Table 5.3). The annuity formula is just a shortcut for discounting each cash flow one by one: $1{,}000/1.07 + 1{,}000/1.07^2 + \ldots + 1{,}000/1.07^5 = 934.58 + 873.44 + 816.30 + 762.90 + 712.99 =$ **4,100.20**. You must use this method when the amounts are *not equal*.

Mixed stream. Example: $400, $800, $500, $400, $300 at the end of years 1–5, discounted at 9%. There is no annuity formula, so discount each one:

Year	Cash flow	÷ 1.09^n	PV
1	400	1.0900	366.97
2	800	1.1881	673.34

3	500	1.2950	386.09
4	400	1.4116	283.37
5	300	1.5386	194.98
Total PV			**1,904.76**

Answering the opening question ($15,000 vs $17,000)
Investing $15,000 today returns $17,000 in 5 years. The implied return = $(17{,}000/15{,}000)^{1/5} - 1 \approx$ **2.53% a year**. If our opportunity cost is 3%, the PV of $17,000 = $17{,}000 \div 1.03^5 \approx \$14{,}664 < \$15{,}000 \rightarrow$ **not worth it**. If the opportunity cost is only 2%, PV ≈ $15,397 > $15,000 → worth it. So the answer depends on the discount rate, not on "17,000 is bigger than 15,000".

Key takeaways

1. Money now > money later; compare at one point in time.
2. $FV = PV(1+r)^n$; $PV = FV \div (1+r)^n$.
3. Discount rate = opportunity cost, reflecting project risk.
4. Annuity due > ordinary annuity (× (1+r)).
5. Perpetuity: CF ÷ r; growing: CF ÷ (r – g).
6. Compare rates using EAR, not nominal rates.
7. Largest nominal total ≠ largest PV; delay destroys value.

Case studies

Case & worked answers

TVM I case: Project Garuda vs Project Samudra — worked solution

Setup: NDI must choose Garuda (11%) or Samudra (12.5%); opportunity cost 10.5%; Rp bn.

A: Q1 FV of 320 at 10.5%: 431.8 / 527.2 / 868.5. Q2 PV of 1,050 in year 10: 443.5 (9%), 369.8 (11%), 283.2 (14%). Q3 PV(G1) = 486.5 > 475 → G2 cheaper; implied rate 13.7%. Q4 equivalent year-5 payment 591.4 → 550 too low.

B: Q5 implied appreciation 7.9% vs 10.5% alternative (205 grows to 337.7 > 300) → sell; 125 is sunk; 205 is the relevant cost. Q6 quarterly 10% wins (EAR 10.38%, Rp163.86).

C: NPV Garuda 259.8 (PV in 1,516.9, out 1,257.1); Samudra 311.5 (PV in 1,412.4, out 1,100.9); nominal net cash 1,620 vs 1,940. Using 11% for Samudra (424.0) overstates it.

D: two-year delay costs Garuda ~48.9; break-even terminal value ~312.3; sensitivities Garuda 418.1/259.8/126.6/14.1, Samudra 507.7/311.5/153.6/49.5. Shocks A/B/C: Samudra stays ahead (182.5/154.6/216.6 vs 126.6/109.5/130.4).

E: financially Samudra leads and leaves more cash headroom (Garuda's 320 upfront would breach the Rp250 minimum if funded from cash alone); strategically weigh domestic predictability vs regional exposure and concentration risk. Separate financial evidence from strategy.

Worked examples

Problem 1 Easy **Future value**

Rp10 juta ditabung 5 tahun dengan bunga 8% per tahun (majemuk tahunan). Berapa nilainya nanti?

1. $FV = PV \times (1 + r)^n = 10 \times 1{,}08^5$
2. $1{,}08^5 = 1{,}4693$.
3. $FV = 10 \times 1{,}4693 = 14{,}69$.

Answer: ± Rp14,69 juta.

Problem 2 Easy **Present value**

Kamu akan menerima Rp50 juta 4 tahun lagi. Dengan tingkat diskonto 10%, berapa nilainya hari ini?

1. $PV = FV \div (1 + r)^n = 50 \div 1{,}1^4$
2. $1{,}1^4 = 1{,}4641 \rightarrow PV = 50 \div 1{,}4641 = 34{,}15$.

Answer: ± Rp34,15 juta.

Problem 3 Medium **Anuitas biasa vs di muka**

Menabung Rp12 juta per tahun selama 10 tahun, bunga 7%. Berapa saldo akhir jika setoran di akhir tahun (anuitas biasa)? Bagaimana jika di awal tahun?

1. $FVA = CF \times [(1 + r)^n - 1] \div r = 12 \times (1{,}07^{10} - 1) \div 0{,}07$
2. $1{,}07^{10} = 1{,}9672 \rightarrow$ faktor = 0,9672 ÷ 0,07 = 13,816.
3. FVA = 12 × 13,816 = 165,8.
4. Anuitas di muka = 165,8 × 1,07 = 177,4 (setiap setoran berbunga satu tahun lebih lama).

Answer: Akhir tahun ± Rp165,8 juta; awal tahun ± Rp177,4 juta.

Problem 4 Easy **Perpetuity and growing perpetuity**

An endowment pays IDR 12 million a year forever, starting next year. Discount rate 8%. (a) What is it worth today? (b) What if payments grow 3% a year (next year still IDR 12 million)?

1. `PV of perpetuity = CF₁ ÷ r = 12 ÷ 0.08 = IDR 150 million`

2. `PV of growing perpetuity = CF₁ ÷ (r - g) = 12 ÷ (0.08 - 0.03) = IDR 240 million`

3. The formula only works if r > g. The closer g gets to r, the more the value explodes — so growth assumptions need care.

Answer: (a) IDR 150 million; (b) IDR 240 million.

Problem 5 Medium **PV of a mixed stream**

A project pays IDR 30 million (end of year 1), IDR 50 million (year 2) and IDR 40 million (year 3). Discount rate 10%. What is its PV?

1. The amounts differ, so discount each one: $PV = CF_t \div (1+r)^t$.

2. $30 \div 1.1 = 27.27$; $50 \div 1.21 = 41.32$; $40 \div 1.331 = 30.05$.

3. Total PV = 27.27 + 41.32 + 30.05 = **98.65**.

Answer: PV ≈ IDR 98.65 million.

Problem 6 Challenging **Solving for the rate and the number of periods**

(a) IDR 20 million grows to IDR 32 million in 6 years. What is the annual rate? (b) At 9%, how many years does money take to double? Compare with the Rule of 72.

1. From $FV = PV(1 + r)^n$:

$$r = (FV \div PV)^{1/n} - 1 = (32 \div 20)^{1/6} - 1 = 1.6^{1/6} - 1 \approx 8.15\%$$

2. $$n = \ln(FV \div PV) \div \ln(1 + r) = \ln 2 \div \ln 1.09 \approx 8.04 \text{ years}$$

3. Rule of 72: 72 ÷ 9 = 8 years — very close to the exact 8.04.

Answer: (a) about 8.15% a year; (b) about 8 years.

Practice questions

Try answering first. The answer key is below each question.

1. FV of 320 at 10.5% for 3 years ≈ …

A. 353.6
B. 390.7
C. 431.8
D. 527.2

Answer: C. 320×1.105^3.

2. If the discount rate rises, PV of future cash…

A. Rises
B. Falls
C. Stays
D. Can't tell

Answer: B. $PV = FV/(1+r)^n$.

3. By the Rule of 72, money at 8% doubles in about…

A. 5
B. 7
C. 9
D. 12

Answer: C. 72 ÷ 8 = 9.

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4. G1 wants 540 next year; G2 475 today. Borrowing cost 8.75%. Choose…

A. G1
B. G2 (G1 implies 13.7%)
C. Same
D. Inflation only

Answer: B. Deferring = borrowing at 13.7%.

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5. An annuity due is worth more than an ordinary annuity because…

A. More payments
B. Each payment arrives one period earlier
C. Higher rate
D. No tax

Answer: B. × (1 + r).

Sources

- -P3MD- Week 6 - Time Value of Money_AZ.pdf (Zutter & Smart bab 5)
- MBA_TVM_I_Redesigned_Visual_Student_Case.docx